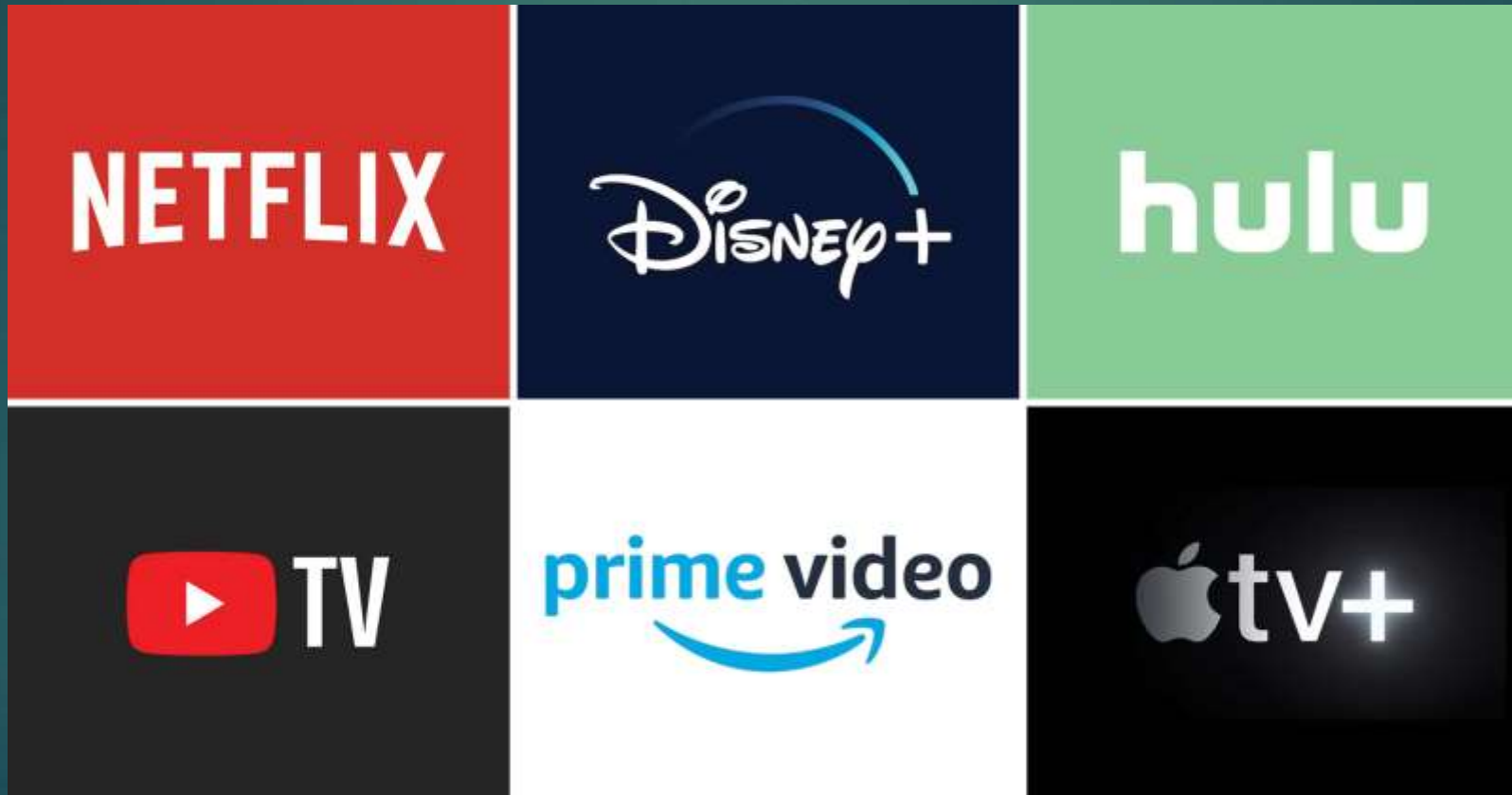


# MOVIE RECOMMENDATION SYSTEM



# OVERVIEW

- ▶ This dataset (ml-latest-small) describes 5-star rating and free-text tagging activity from [MovieLens](<http://movielens.org>), a movie recommendation service. It contains 100836 ratings and 3683 tag applications across 9742 movies. These data were created by 610 users between March 29, 1996 and September 24, 2018. This dataset was generated on September 26, 2018.
- ▶ Users were selected at random for inclusion. All selected users had rated at least 20 movies. No demographic information is included. Each user is represented by an id, and no other information is provided.
- ▶ The data are contained in the files `links.csv`, `movies.csv`, `ratings.csv` and `tags.csv`. More details about the contents and use of all these files follows.

# OBJECTIVE

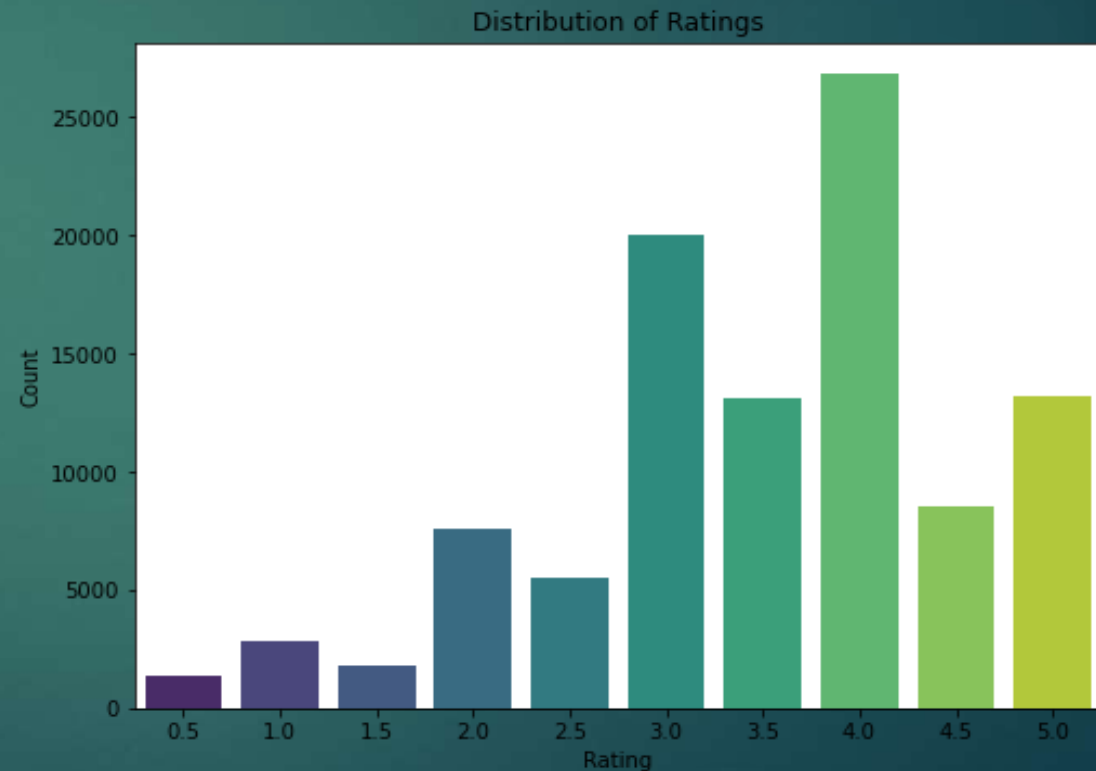
- ▶ The objective of this project is to build a recommendation system that suggests the top 5 movies to a user based on their past ratings. This will simulate a real-world scenario where a platform wants to improve user satisfaction by offering tailored recommendations.

# DATA UNDERSTANDING

- ▶ This dataset (ml-latest-small) describes 5-star rating and free-text tagging activity from [MovieLens](<http://movielens.org>), a movie recommendation service. It contains 100836 ratings and 3683 tag applications across 9742 movies. These data were created by 610 users between March 29, 1996 and September 24, 2018. This dataset was generated on September 26, 2018.
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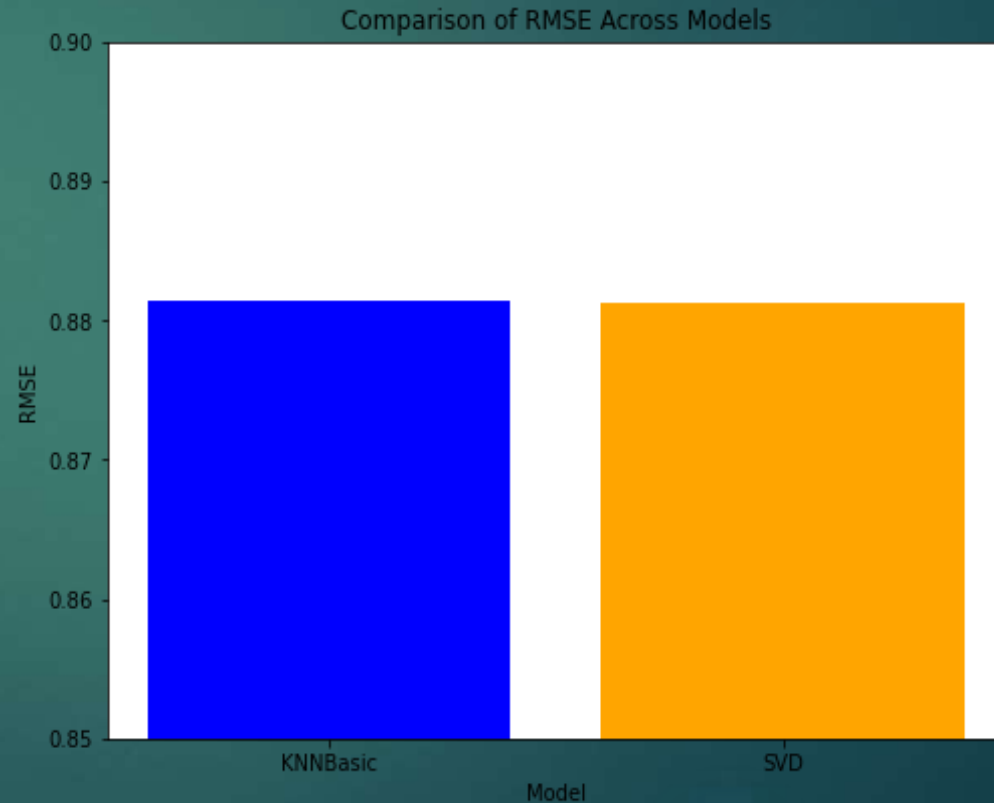
# DISTRIBUTION OF RATINGS

- ▶ Ratings are distributed across all possible values(1-5), with most ratings being 4 or 5.



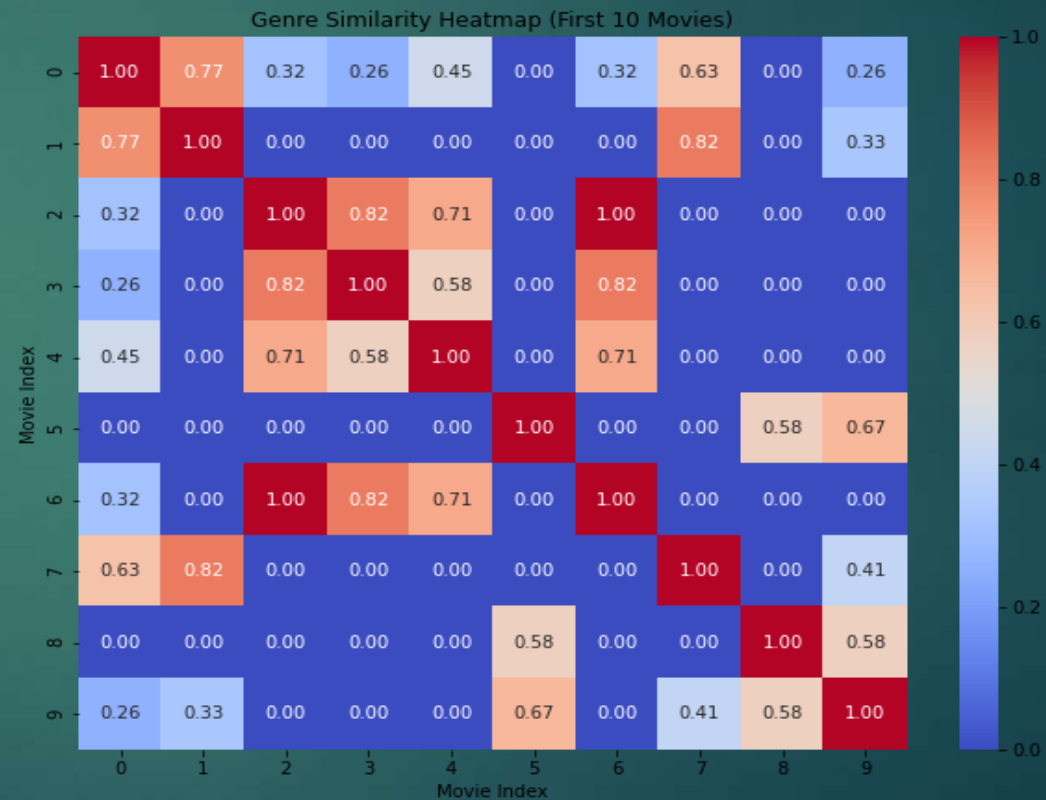
# COMPARISON OF RMSE ACROSS THE MODELS

- ▶ - Both models are effective at predicting user ratings, as indicated by their low RMSE values.
- ▶ - The slight edge of the SVD model could be due to its ability to capture latent factors in the data more effectively than KNNBasic.

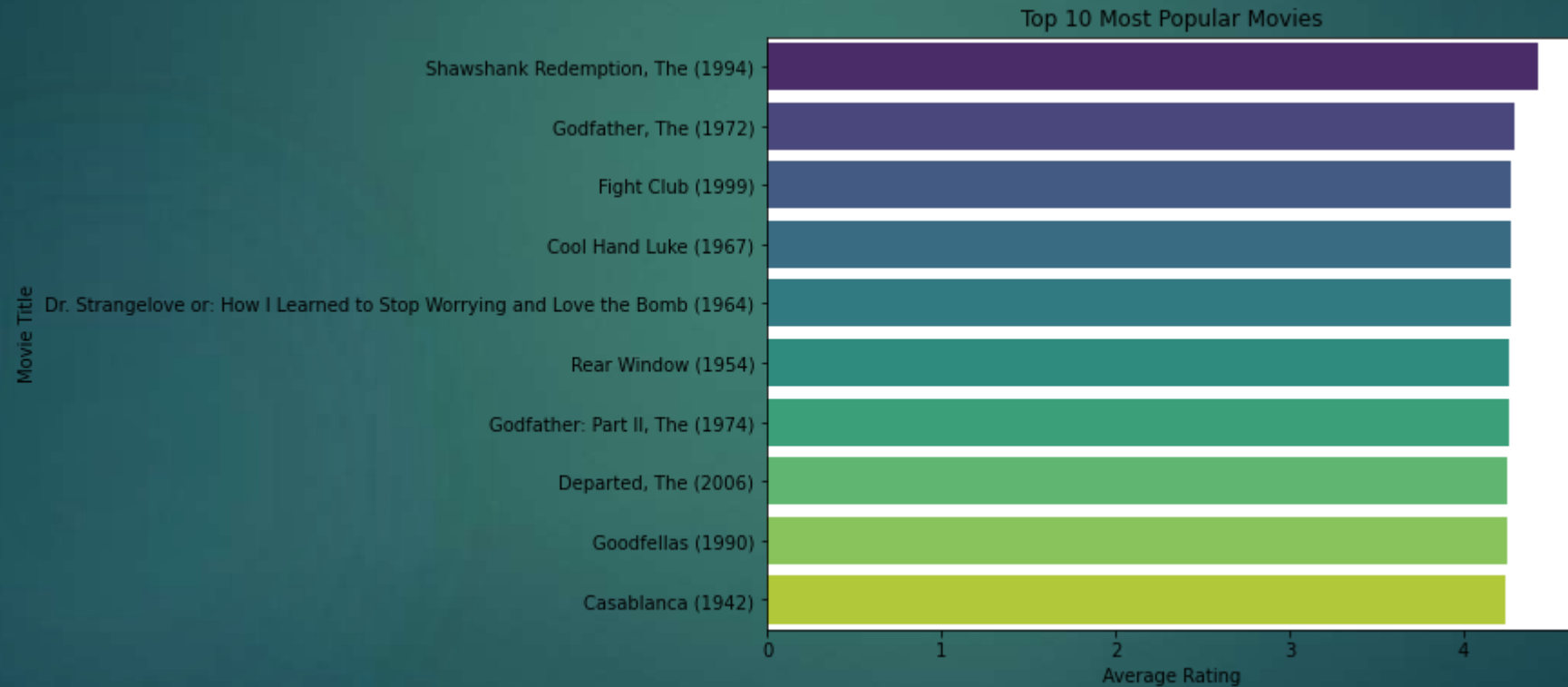


# GENRE SIMILARITY MATRIX

- ▶ High similarities (red cells) indicate that the items are very similar to each other.
- ▶ Low similarities (blue cells) indicate that the items are dissimilar.
- ▶ Moderate similarities (orange/yellow cells) indicate some degree of similarity.

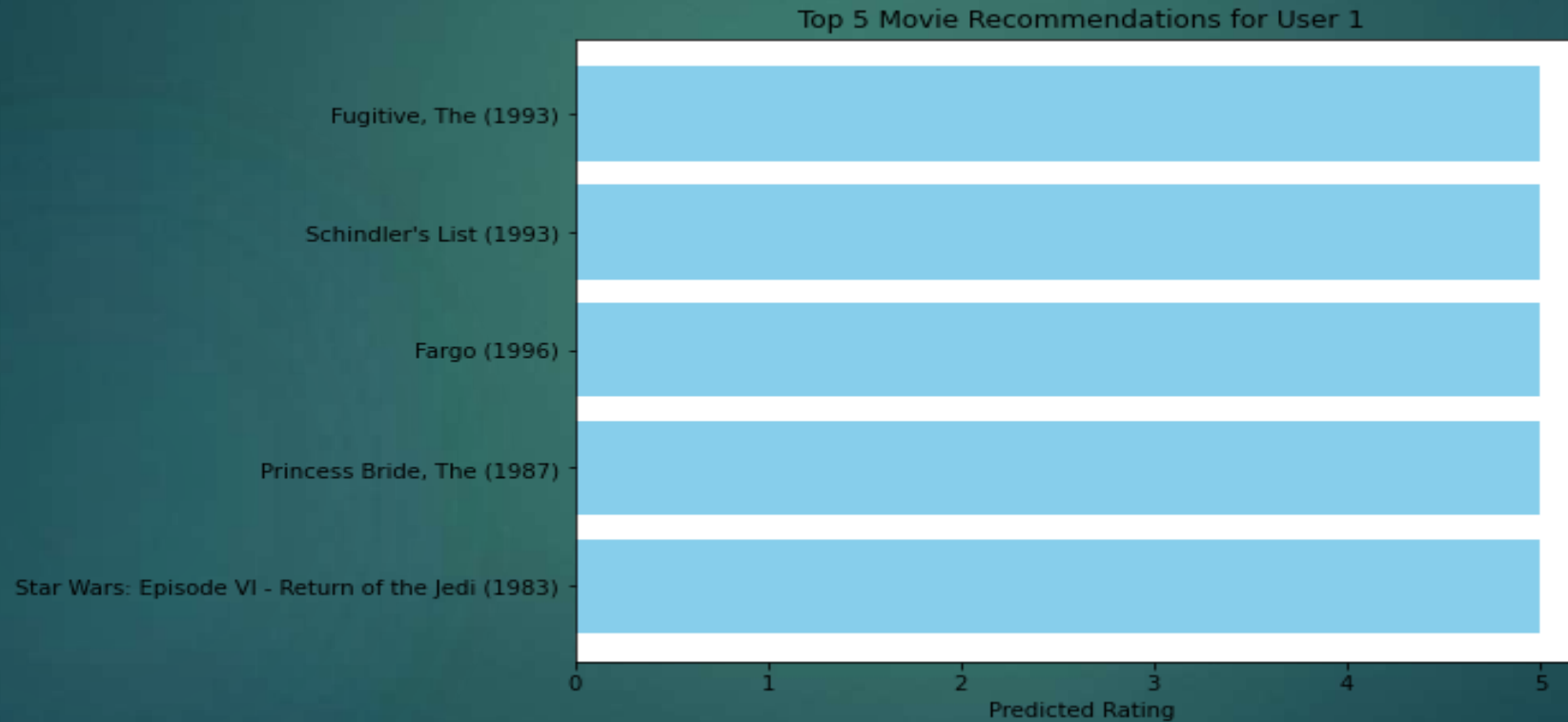


# VISUALISATION OF THE MOVIES WITH THE HIGHEST AVERAGE RATINGS





# TOP 5 RECOMMENDATIONS FOR USERS



# RECOMMENDATIONS

- ▶ Incorporate content-based filtering to address the cold-start problem.
- ▶ Experiment with advanced techniques like matrix factorization (SVD) or neural networks.
- ▶ Deploy the model as a web application for real-time recommendations.